Functional Genomic Technologies Assignment

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Introduction to the Data

The data I was assigned to use in this assignment were the data that was generated for the paper “WBSCR16 is essential for mitochondrial 16S rRNA processing in mammals” by Zhang et al. (2025) (1). It has only one condition which is genotype. There are 6 samples: 3 of them are floxed genotype and 3 of them are *WBSCR16* Knock-outs (1). The reads were submitted to GEO database with accession number GSE229693 (1,2). In this paper, floxed genotype is annotated as “FL” while knock-out genotype is annotated as “KO”.

Outline

- I performed the quality analysis on the raw reads using code that was given in lecture (3).
- Later, I performed another quality analysis on aligned reads (3).
- Using the information from GEO database (1,2), I created an annotation file to use for RNA-Seq analysis.
- I used the annotation file and the table containing the aligned reads and counts for the RNA-Seq analysis on R (4) using the pipeline given in Functional Genomic Technologies class material (Week 6 (5) and Example Analysis (6)).

PART A: Quality Analysis

I analysed the quality of the samples with Multiqc (version 1.22.3)(7) software. Overall, the quality was good. The alignment scores for each sample were in range of 94.6% to 93.3%. There were 6 overrepresented sequences, two of them occurring in 5 of the samples, and other two occurring in 4 of the samples. I put the sequences on BLAST (8) to understand their sources. All the sequences belonged to *Mus musculus* and were either mRNA or cDNA (9). The data did not need trimming. It was 150bp in length. The duplication levels were high.

PART B: RNA-Seq Analysis

I performed the analysis with R (version 4.4.3 (2025-02-28) -- "Trophy Case") (4) on RStudio (version 2024.9.1.394-Cranberry Hibiscus) (10) using both lecture materials provided by the lecturer (5,6). First, I loaded the data on R, the packages that I used are as follows: Rsubread(11) , edgeR (12), limma (13), gplots (14), DESeq2 (15), affy (16), QuasR (17), pheatmap (18) and EnchancedVolcano (19). After loading the data into DeSeq2 (15), I looked into boxplots of normalised counts which could be seen in Figure 1.

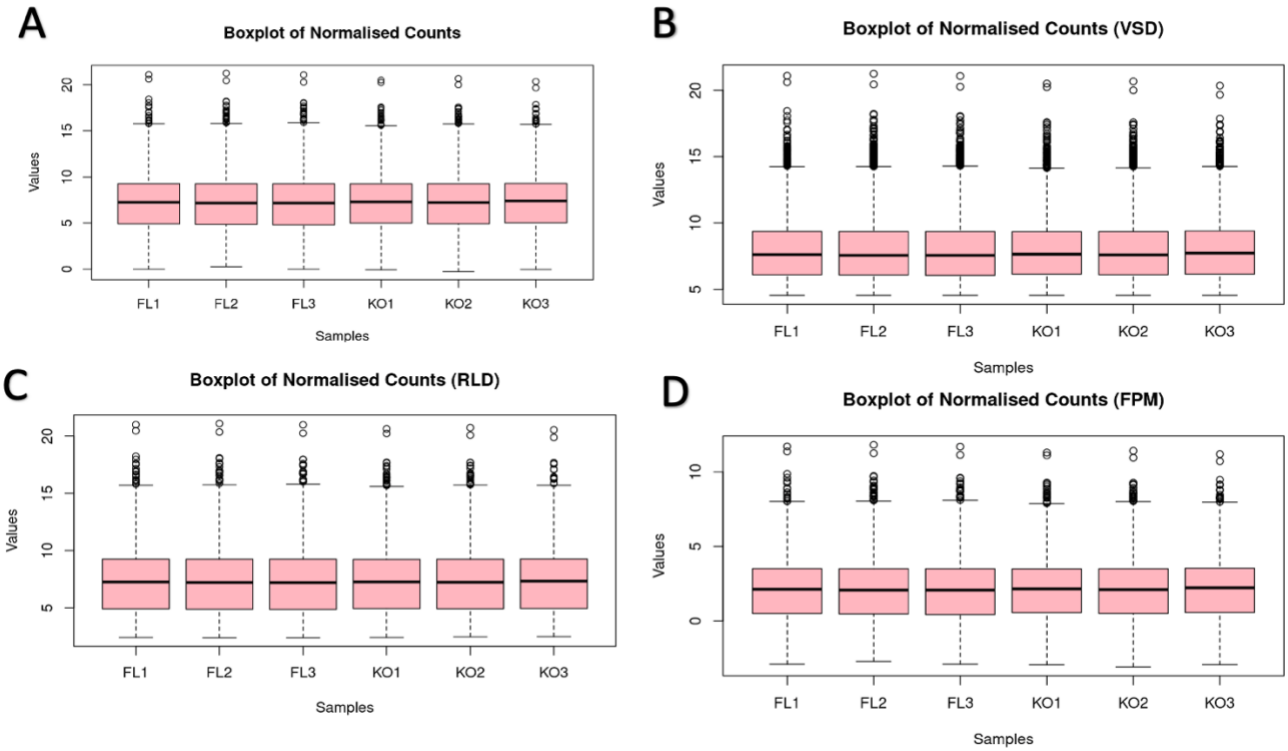


Figure 1: Boxplots of normalised counts. A: Normalised counts on log2 scale. B: Normalised counts on VST scale. C: Normalised counts on RLD scale. D: Normalised counts on FPM scale.

I have tried to understand the differences between the methods of normalisation therefore in Figure 1, there are several boxplots according to the normalisation technique. Mainly, they all look similar except for FPM in which the values seem to be lower. As much as I can tell, the outliers make the difference between the methods. The difference in outliers between RLD and VST could be the result of size factors (15). With normalisation, the boxplots seem to align which means that normalisation was successful (20).

Then, I looked at the data without normalisation.

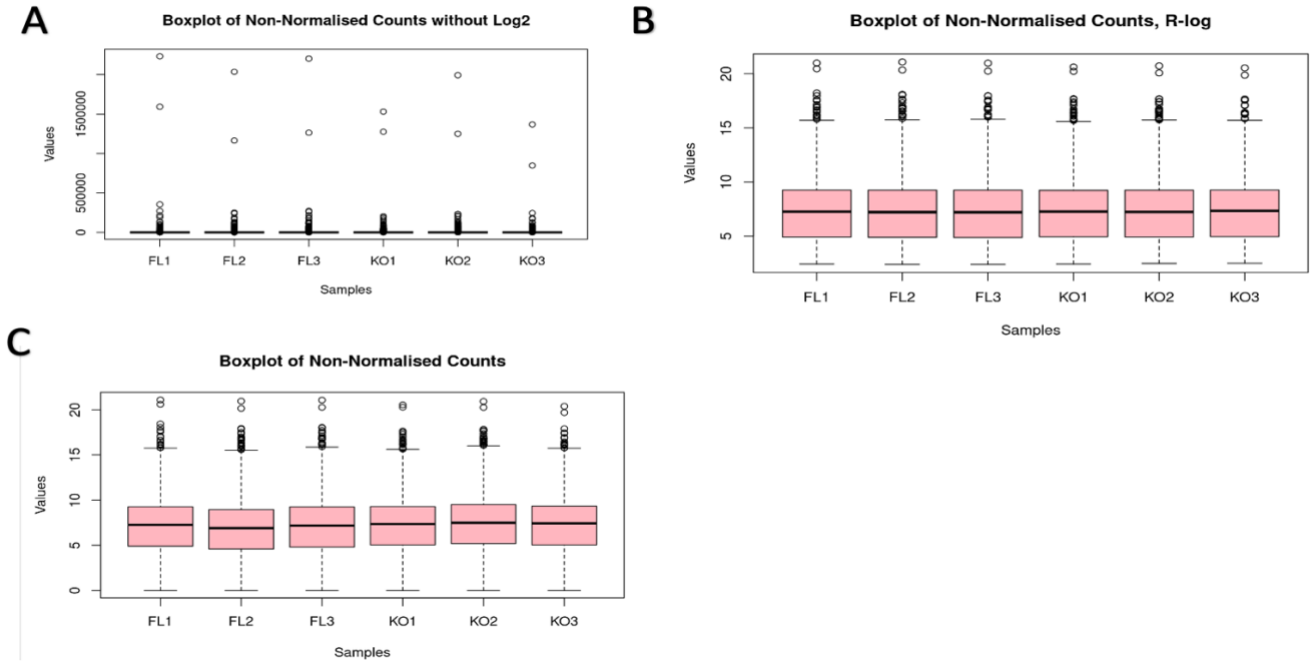


Figure 2: Boxplot of non-normalised counts. A: Non-normalised counts without a scale. B: Non-normalised counts on log2 scale. C: Non-normalised counts on RLog scale.

Without a scale, it is very difficult to understand the changes and outliers of the counts as seen in Figure 2A. Rlog and log2 transformation of non-normalised counts do not seem to differ in a major way.

After this, I performed PCA (Principal Component Analysis) plots to understand the data and its character further (21).

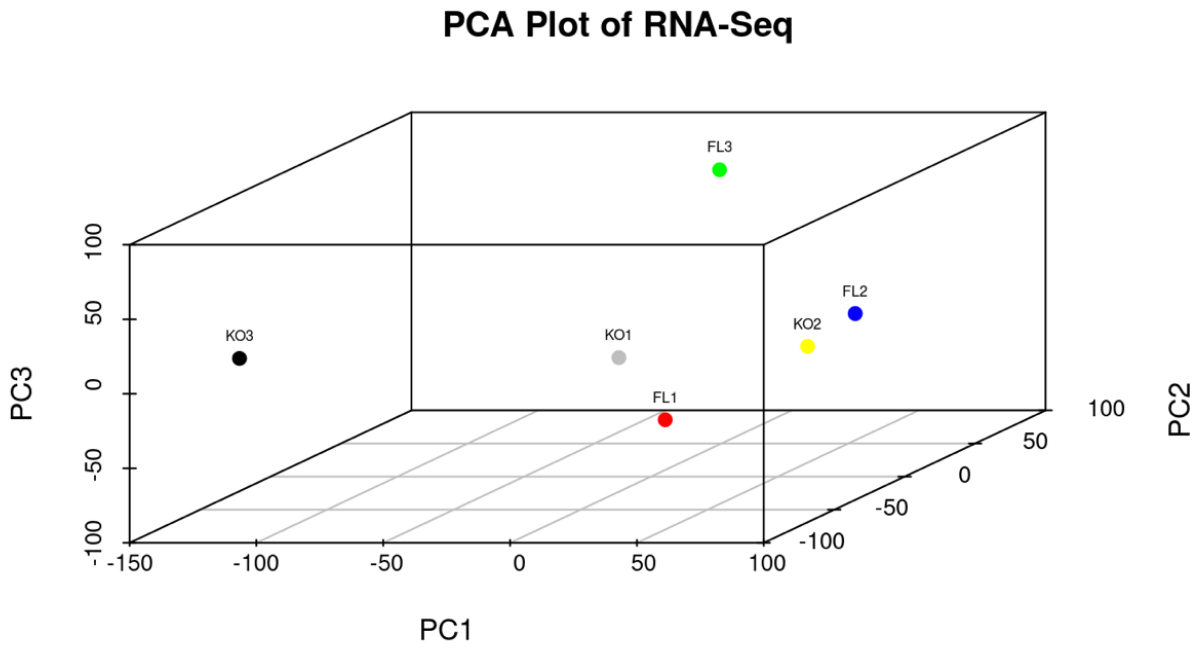


Figure 3: 3D PCA analysis of normalised counts.

From this figure, I can see that while KO1 and KO2 are similar, KO3 and FL3 are very distant to other samples. This is interesting as none of the FL samples seem to be similar to each other. This could mean that the organisms that were used in this experiment were from heterogenous populations (21). After that, I plotted the PCA values on 2D Plots which can be seen in Figure 4 and 5.

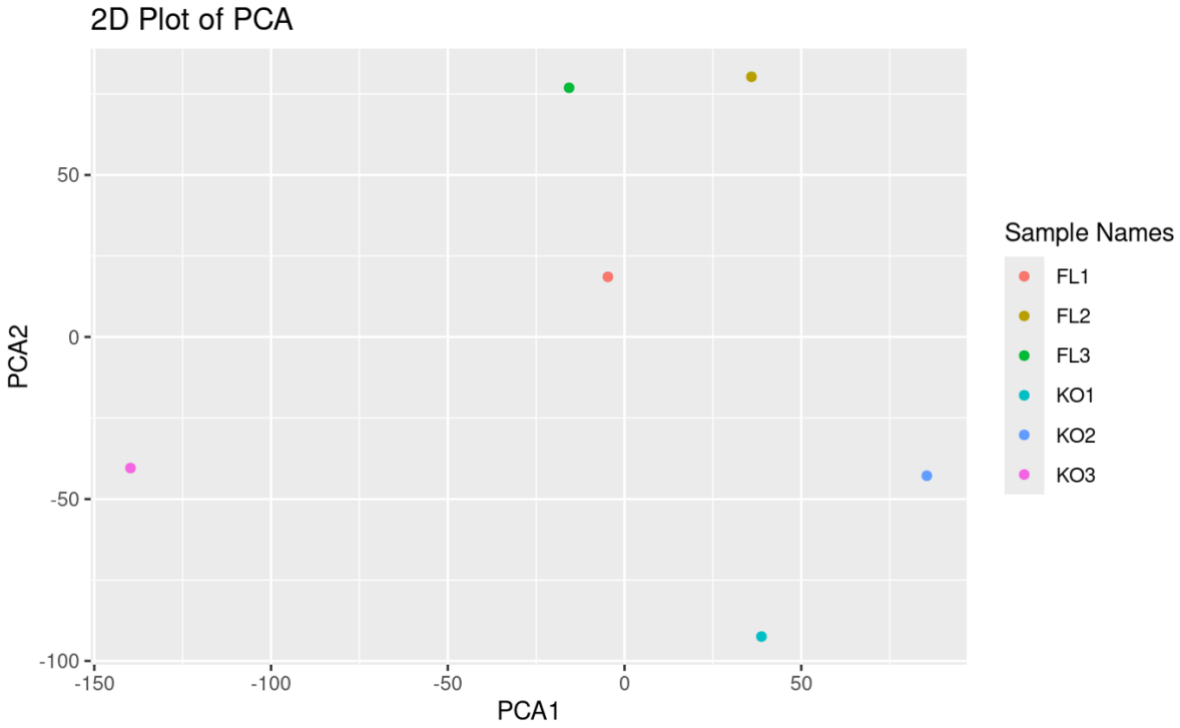


Figure 4: 2D Plot of PCA on normalised counts according to samples.

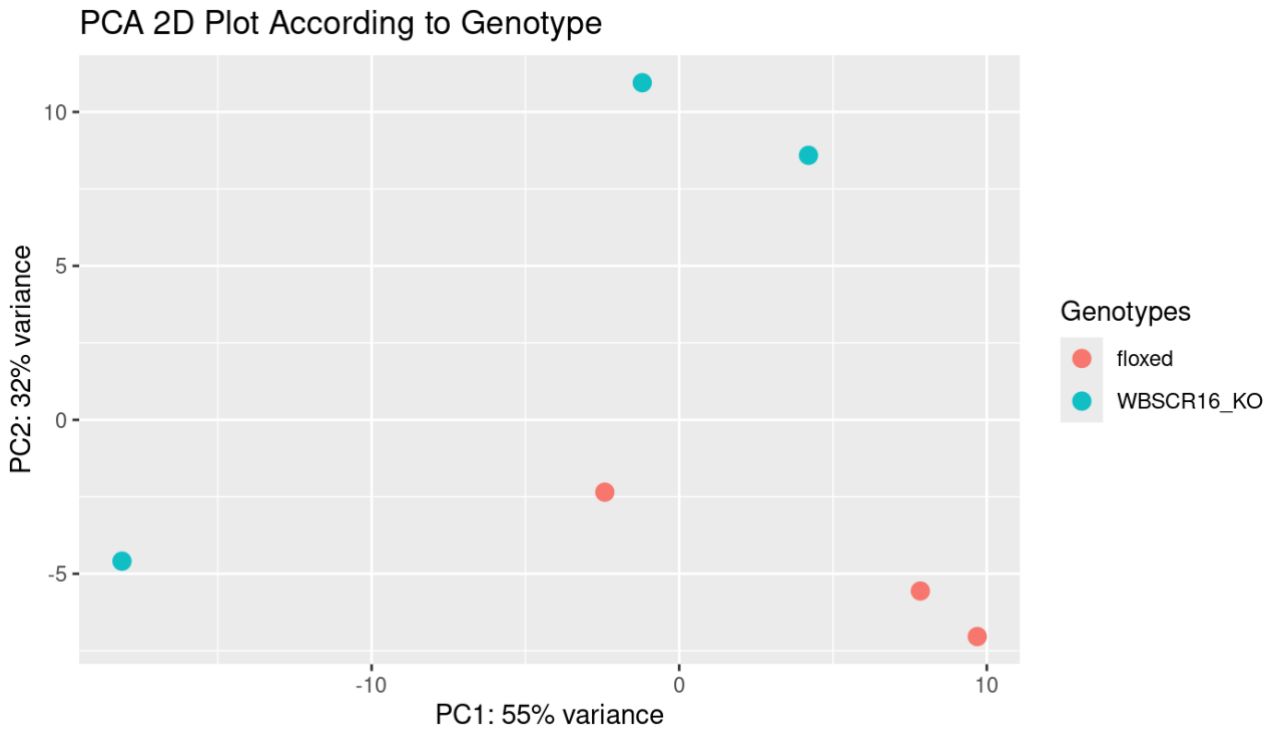


Figure 5: 2D Plot of PCA on normalised counts according to genotype.

In Figure 4 and 5, I can see that FL samples are closer to each other which is not the case for KO samples because KO3 is further away from the rest of the samples. Then, I decided to analyse the similarities of the samples according to PCA further and plotted a heatmap for the values.

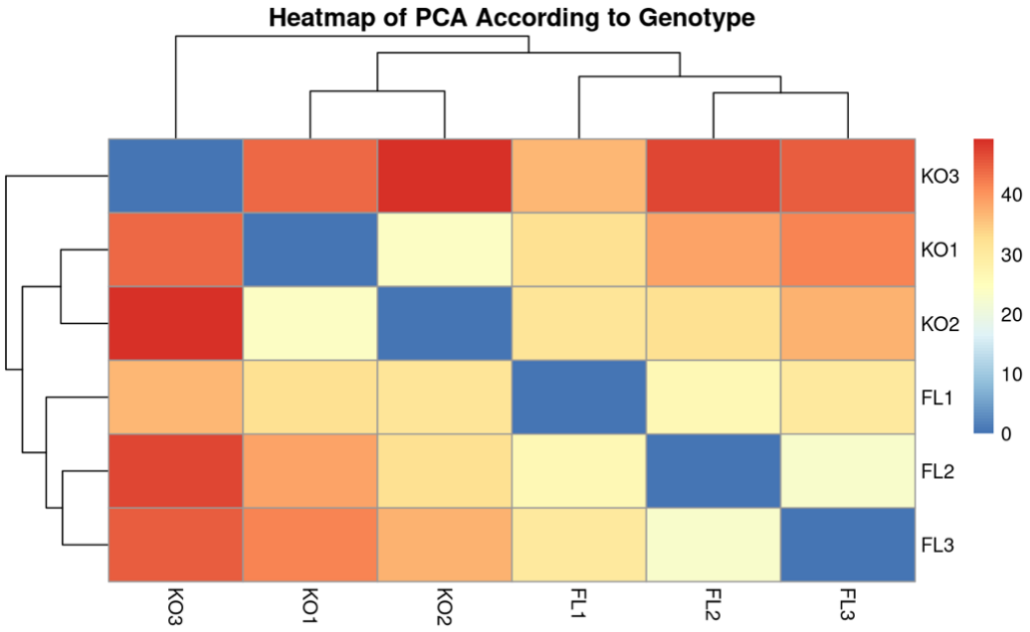


Figure 6: Heatmap of PCA Analysis according to genotype.

In Figure 6, I can see that samples are not that dissimilar to each other except for KO3, which does not seem to be similar to any other sample (closest being FL1). What I find the most interesting is that KO3 is more distant to KO2 than FL genotypes. However, since the only difference between samples is knocking out of a gene, it is possible that the rest of the expression pattern is closer to the other samples.

After this, I wanted to see the top 50 genes with highest values of RLD.

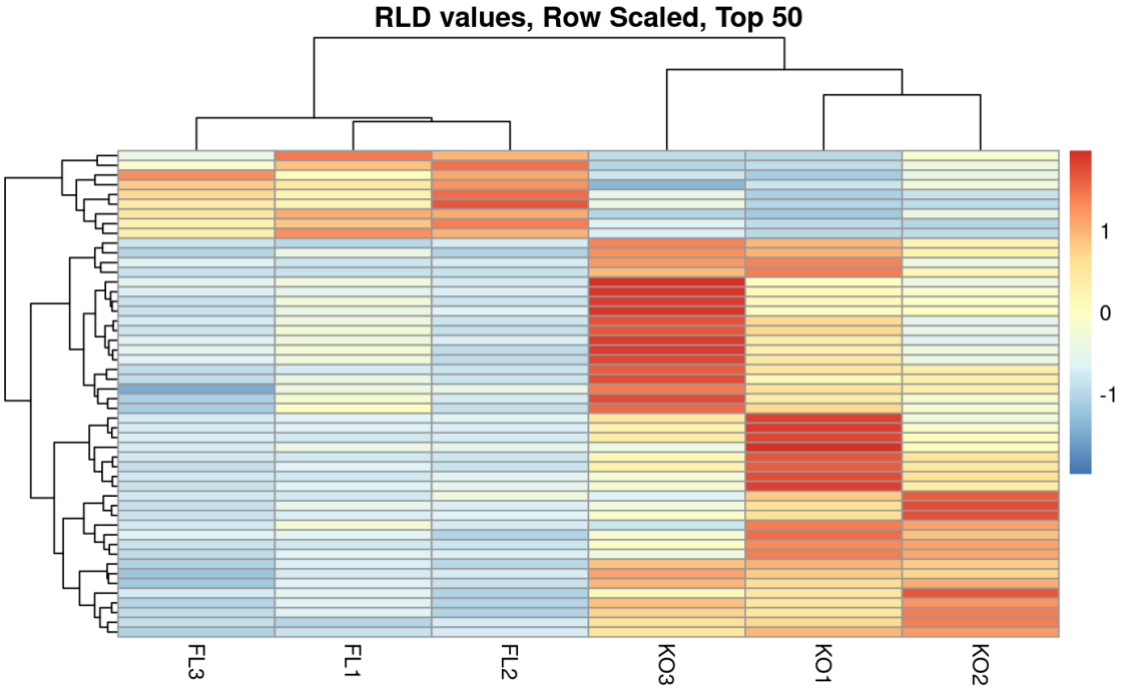


Figure 6: Heatmap of RLD values of the samples with top 50 genes selected.

I can see that the knockout samples had genes with higher RLD values. This made me curious as I wondered which genes are highly expressed in knockout samples compared to the floxed samples as it could be related to WBSCR16 function. To understand if that was the case, I decided to select the genotype and plot the heatmap again.

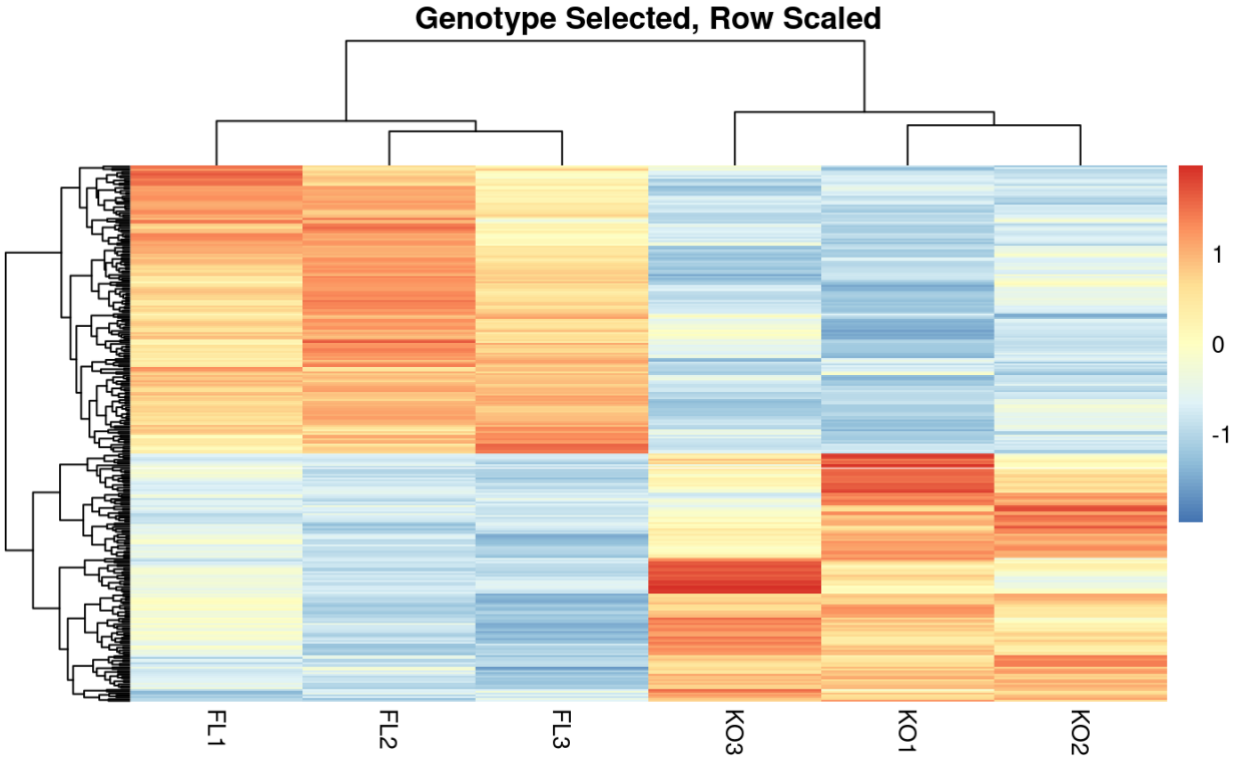


Figure 7: Heatmap of RLD values of the samples with top 50 genes, genotype selected.

Figure 7 has shown to me that the genotypes rather had different genes with high RLD values, however, as I can see from the figure, the floxed genotype did not have genes with values as high as knockouts, especially KO3. This made me curious, and I wanted to see which genes were top 10, and what was their function. This is an important point for me as it could be a topic for another research.

According to this, I plotted the top 10 genes.

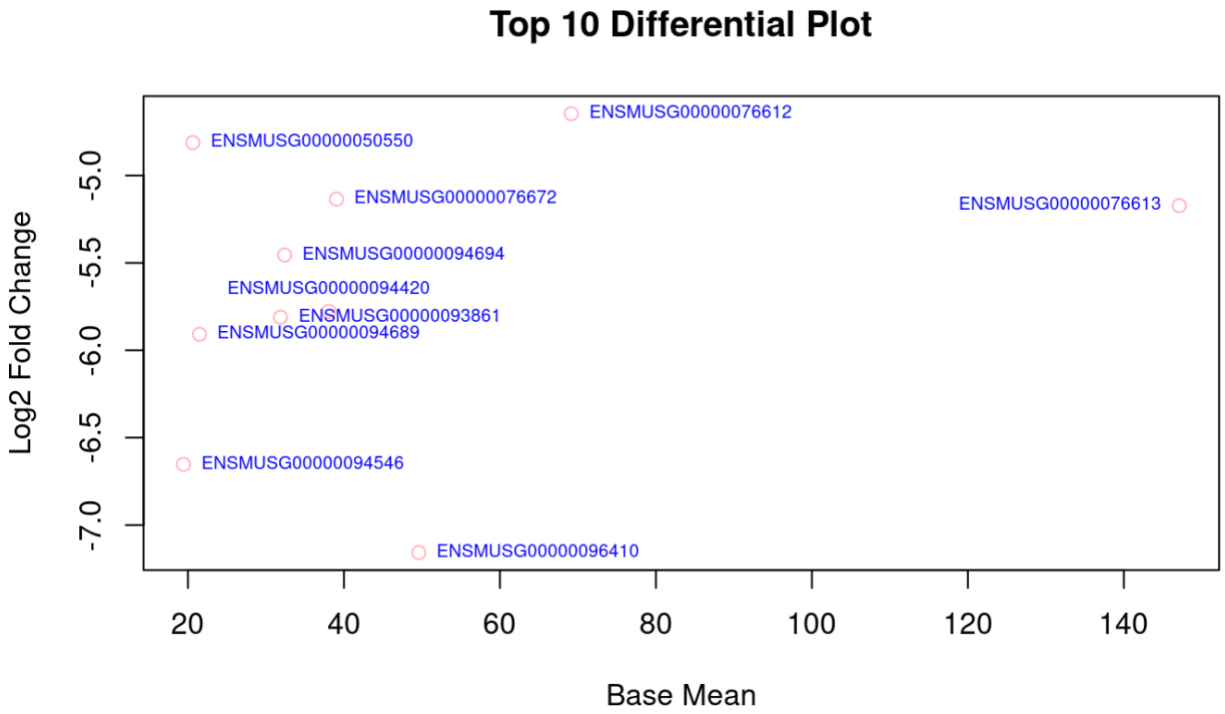


Figure 8: Differential Expression Plot of Top 10 genes.

I looked up the names of the genes on this plot to see if they are mentioned in the paper which they were not. Then, I searched for each gene on NCBI Gene Database (9). In 10 genes, 8 of them were associated with immunoglobulin (9). Hence, I wondered if the *WBSCR16* gene or its products are related to the immune system or perhaps the mouses had an ongoing infection.

I followed this analysis with dispersion plot.

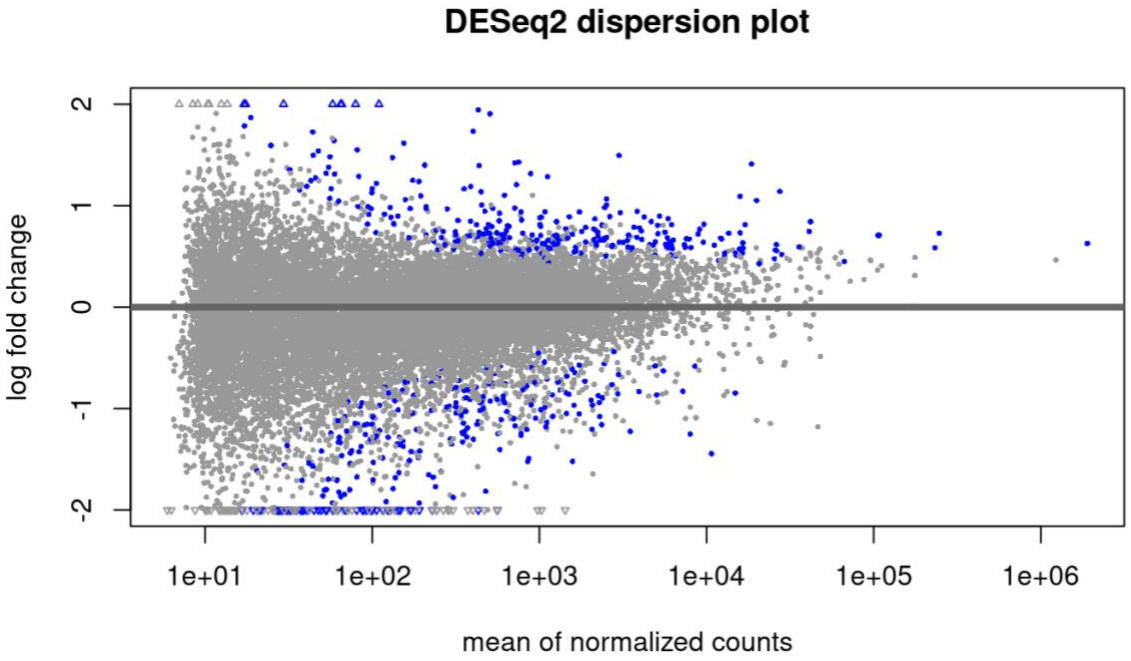


Figure 9: Dispersion plot of the counts against their fold change.

The dispersion plot has shown that there are a lot of genes that are highly expressed according to their fold change (22). More importantly, the fold change for numerous genes is in negatives. Mainly, the fold change values are either 0 or close to zero which means most of the genes did not experience a change in expression (15).

After that I looked into the first 50 genes as candidate genes in a heatmap. The results I obtained are in Figure 9.

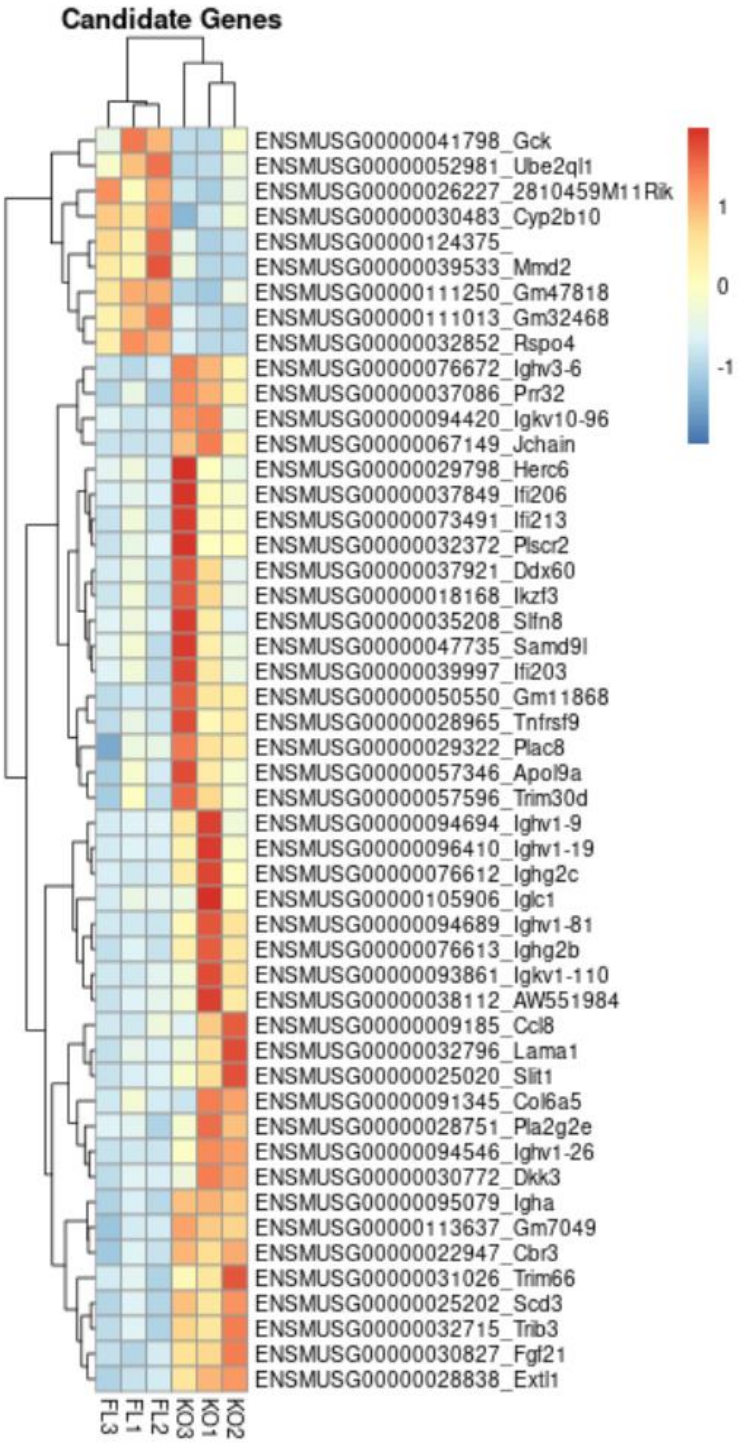


Figure 9: Heatmap of first 50 genes.

According to Figure 9, I see that first 9 genes are more specific for floxed while rest of them are more specific to knock-out genes. This could indicate that first 9 genes are more dependent on *WBSCR16* while the rest are repressed by it directly or indirectly. As shown in the figure, each knockout sample have high expression of different genes, KO3 has the most genes with highest change in expression however none of those genes seem to show the same patterns in other KO samples. This could be one of the reasons why we see in PCA analysis which indicated KO3 had a very different characteristics compared to other samples.

Moreover, when looked into KO1, I can see the genes associated with immunoglobulin are all highlighted which is not the case for other samples. This made me lean more towards my hypothesis of the mouse being sick or having an infection at the time of the study as the other knock-out samples or floxed samples did not show the same pattern of expression. The result of this graph is fascinating as it shows how much knockout samples differ without having nearly nothing in common. This also makes it hard to talk about the direct effect of *WBSCR16* on the genes.

The paper also talked about fatty acid metabolism, and they found that several genes in the pathway were affected by the knockout of *WBSCR16* (1). I could not verify this finding from my analysis as I could only see Sdc3 from the listed genes in Figure 9 and it did not have high values compared to the rest of the genes that can be seen on the heatmap.

One of the genes did not have annotation on them (ENSMUSG00000124375) which caught my attention and I searched for it on the NCBI (9) and Ensembl (23) websites. I found out that it is a long non-coding RNA of a novel gene (23).

After this I plotted a volcano plot of the genes that the paper either talked about or the genes I chose to look into according to the western blot experiments on the paper. The information regarding their names and changes I expect to see according to the paper (1) are shown in Table 1. I found the names of the genes from Ensembl (23).

Table 1: Gene names and the reported effects of *WBSCR16* knockout on them according to the paper (1).

Gene Name	Reported Effect in Floxed Genotype	Reported Effect in Knockout Genotype
mt-Rnr2	Unchanged	Downregulated
mt-Rnr1	Unchanged	Unchanged
mt-Co1	Upregulated	Downregulated
Sdhb	Unchanged	Unchanged
Cox4i1	Upregulated	Downregulated
Cox4i2	Upregulated	Downregulated
Mrps16	Upregulated	Downregulated
Mt-Nd3	Was highly upregulated in KO compared to floxed genotype	
Rcc1l (Wbscr16)	Unchanged	Knocked-out

In addition to the genes here, Eral1 was added as it is associated to 12S rRNA (mt-Rnr1) (9)(23).

Then I plotted a volcano plot with these genes labelled.

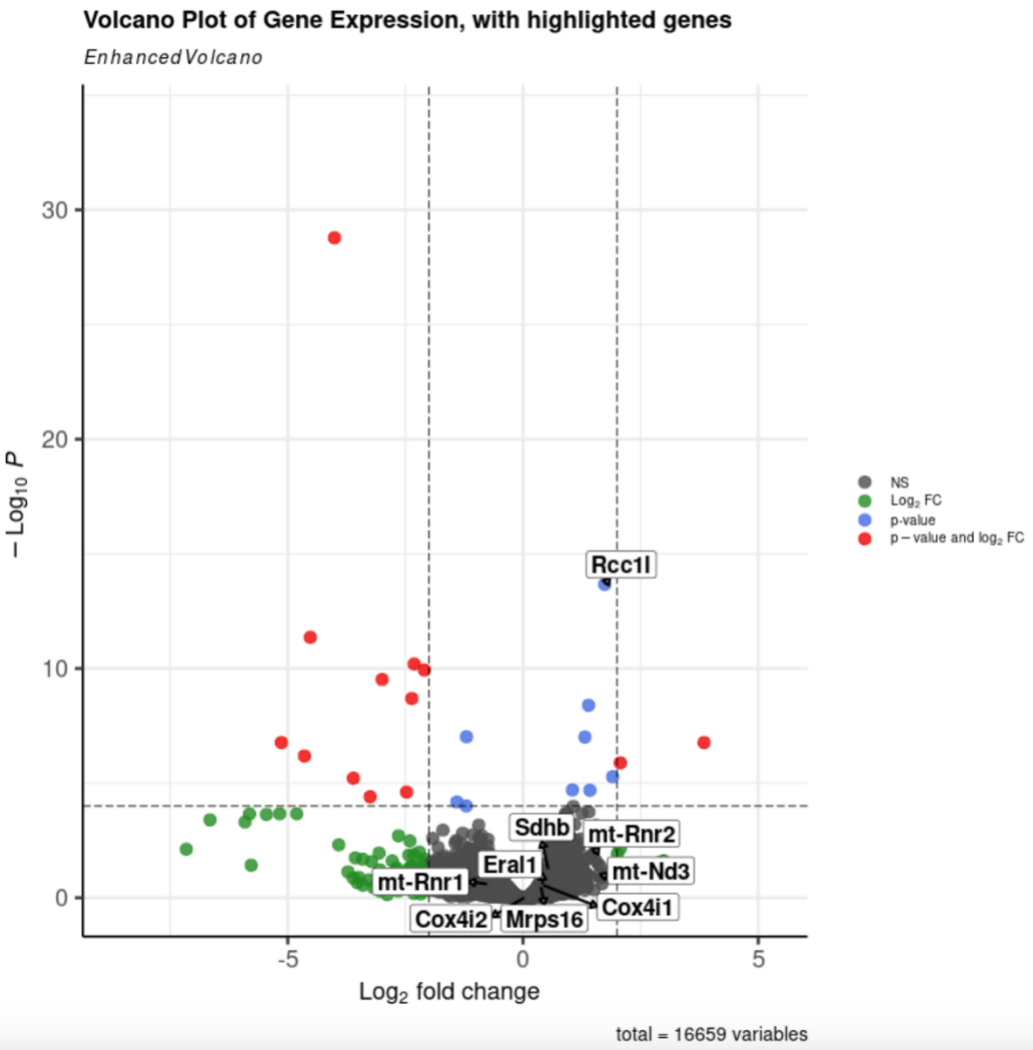


Figure 10: Volcano plot of gene expression fold change.

I expected most of the genes to be around 0 for the fold change as I choose either the ones that had higher expression in floxed and less expression in knockout samples or the ones that remained unchanged across the study to understand how they differ in terms of computational analysis. Moreover, I also chose genes that did not change in floxed genotype but was downregulated in knockouts such as *mt-Rnr2* (1). Therefore, I was expecting to see see a drastic change in mt-Rnr2 which was not the case.

Rccl1 is another name for *Wbscr16*, and as in this study, the gene was knocked out in half of the samples which explains the position of it however, another gene has a higher value for log2FC and p-value which is confusing. It is mentioned in the paper that fatty acid metabolism of knocked out mice were affected and was altered hence, it could be a gene related to that pathway (1) or it could be one of the genes associated with immunoglobulin (sample KO2).

After this, to further understand the fold changes, I decided to plot the genes I wanted to see from the paper (1).

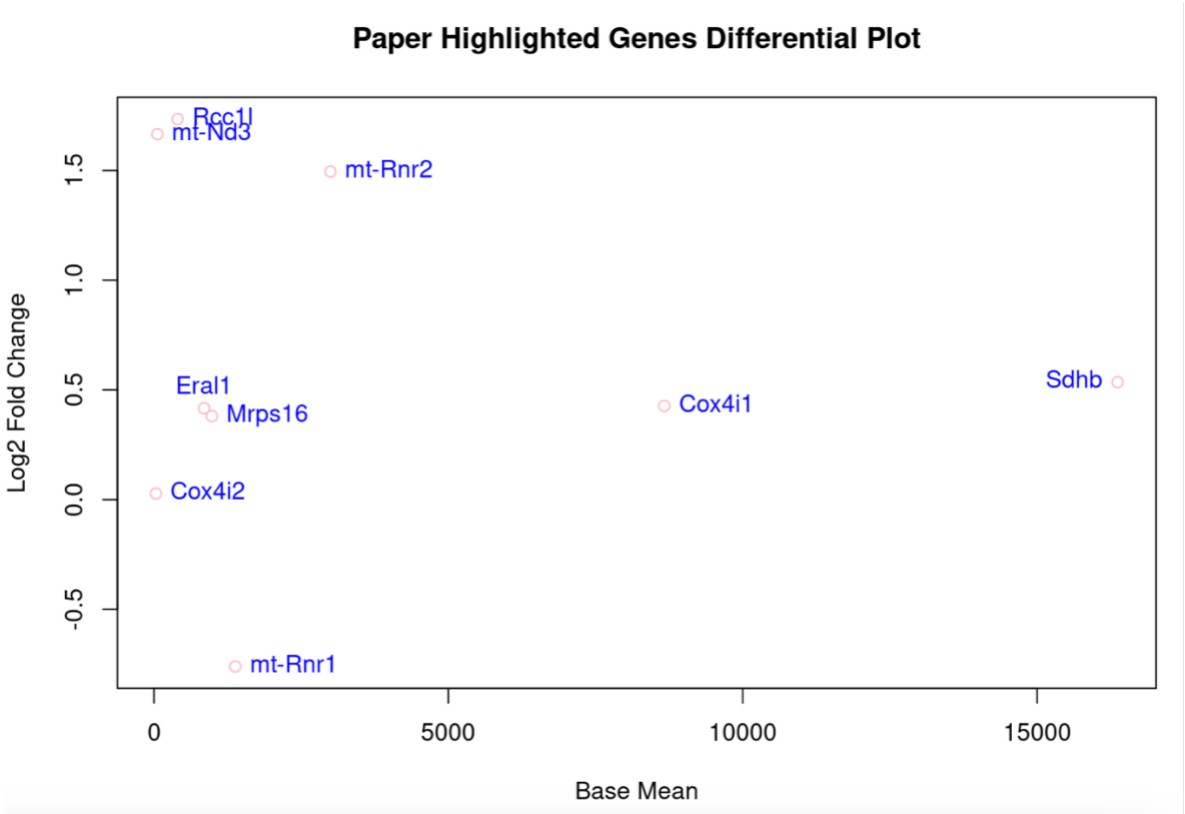


Figure 11: Plot of log2 fold change against mean base for highlighted genes.

In this plot, I can see Rcc1l and mt-Nd3 expressions are differential across the samples which was expected as former is knocked out and the latter was found to have a significant change between genotypes (1). As expected Eral1, Cox4i1, Mrps16 and Sdhb had little changes in expression. Cox4i2 was almost unchanged. According to the paper, 16S rRNA should be downregulated in knockout samples which is annotated as mt-Rnr2 and I can see that from high fold change across the samples as it is supposed to be unchanged for floxed samples (1). Hence, I can see a similar pattern to Rcc1 and mt-Nd3. The paper has stated that mature 12S Rrna was not affected by the knockout (1). To see that, I have included annotations associated with 12S rRNA (mt-Rnr1 and Eral1). However, mt-Rnr1 is the mitochondrial 12S rRNA (9) and it is interesting that it was reported to remain unchanged when it has more than -0.5 fold change.

In conclusion, the RNA-Seq data had good quality and annotation percentage. Additionally, the normalisation methods did not differ a lot from each other. Moreover, PCA analysis showed that the sample KO3 had different characteristics from the rest of the samples which was further proven by the heatmap of top 50 differentially expressed genes. Analysing top 10 differentially expressed genes have given me the information that some of the genes associated with immunoglobulin had high differential expression which was then proved to be coming from the sample KO1 with heatmaps. At last, the findings from the paper could be mostly verified with a question mark on the fold change of mt-Rnr1 which is 12S rRNA.

References

1. Zhang S, Dong Z, Feng Y, Guo W, Zhang C, Shi Y, et al. WBSCR16 is essential for mitochondrial 16S rRNA processing in mammals. Nucleic Acids Res. 2025 Feb 10;53(3).
2. Edgar R, Domrachev M, Lash AE. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. Nucleic Acids Res. 2002;30(1).
3. Tomlinson S. FGT ICA Assessing Sequence and Alignment Quality with MultiQC [unpublished lecture notes]. PGBI110402024: Functional Genomic Technologies, University of Edinburgh; lecture given 2025, March 21.
4. R Core Team (2025). _R: A Language and Environment for Statistical Computing_. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
5. Tomlinson S. FGT T7 Advanced Analysis RNA-Seq Using R [unpublished lecture notes]. PGBI110402024: Functional Genomic Technologies, University of Edinburgh; lecture given 2025, March 3.
6. Tomlinson S. Example Analysis Extra ICA Help [unpublished lecture notes]. PGBI110402024: Functional Genomic Technologies, University of Edinburgh.
7. Ewels P, Magnusson M, Lundin S, Käller M. MultiQC: Summarize analysis results for multiple tools and samples in a single report. Bioinformatics. 2016;32(19).
8. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. Basic local alignment search tool. J Mol Biol. 1990;215(3).
9. Gene [Internet]. Bethesda (MD): National Library of Medicine (US), National Center for Biotechnology Information; 2004 – [cited 2025 04 20]. Available from: https://www.ncbi.nlm.nih.gov/gene/.
10. Posit team. RStudio: Integrated Development Environment for R [Internet]. Boston, MA: Posit Software, PBC; 2024 [cited 2025 Apr 21]. Available from: http://www.posit.co/
11. Liao Y, Smyth GK, Shi W. The R package Rsubread is easier, faster, cheaper and better for alignment and quantification of RNA sequencing reads. Nucleic Acids Res. 2019;47(8).
12. Chen Y, Chen L, Lun ATL, Baldoni PL, Smyth GK. edgeR 4.0: powerful differential analysis of sequencing data with expanded functionality and improved support for small counts and larger datasets. bioRxiv. 2024;
13. Ritchie ME, Phipson B, Wu D, Hu Y, Law CW, Shi W, et al. Limma powers differential expression analyses for RNA-sequencing and microarray studies. Nucleic Acids Res. 2015;43(7).
14. Warnes GR, Bolker B, Bonebakker L, Gentleman R, Liaw WHA, Lumley T, et al. Package “gplots”: Various R programming tools for plotting data. R package version 2170. 2016;(1).
15. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol. 2014;15(12).
16. Gautier L, Cope L, Bolstad BM, Irizarry RA. Affy - Analysis of Affymetrix GeneChip data at the probe level. Bioinformatics. 2004;20(3).
17. Gaidatzis D, Lerch A, Hahne F, Stadler MB. QuasR: Quantification and annotation of short reads in R. Bioinformatics. 2015;31(7).
18. Kolde R. Package `pheatmap’. Bioconductor. 2012;
19. Blighe K, Rana S, Lewis M. Publication-ready volcano plots with enhanced colouring and labeling. Bioconductor. 2022;
20. Lin Y, Golovnina K, Chen ZX, Lee HN, Negron YLS, Sultana H, et al. Comparison of normalization and differential expression analyses using RNA-Seq data from 726 individual Drosophila melanogaster. BMC Genomics. 2016;17(1).
21. Son K, Yu S, Shin W, Han K, Kang K. A simple guideline to assess the characteristics of RNA-Seq Data. Biomed Res Int. 2018;2018.
22. McDermaid A, Monier B, Zhao J, Liu B, Ma Q. Interpretation of differential gene expression results of RNA-seq data: Review and integration. Vol. 20, Briefings in Bioinformatics. 2019.
23. Harrison PW, Amode MR, Austine-Orimoloye O, Azov AG, Barba M, Barnes I, et al. Ensembl 2024. Nucleic Acids Res. 2024;52(D1).